

T10

$$h = 100$$

0	1	2	3	4	5	6	7	8	9	
5	8	6	12	14	18	11	6	13	7	- m;

$$a) H_0: g \sim R(0; 10) \quad H_1: \bar{H}_0 \quad \alpha = 0,05$$

$$p_i = \frac{1}{10} \quad n/p_i = 10$$

$$\tilde{D} = 16,4$$

$$p\text{-value} = P(\tilde{D} \geq 16,4 / H_0) = 0,059 > \alpha \quad \Delta \leadsto \chi^2(9) \rightarrow \text{не отвер. } H_0$$

$$b) H_0: g \sim N(\theta_1; \theta_2^2) \quad p(x) = \frac{1}{\sqrt{2\pi\theta_2^2}} e^{-\frac{(x-\theta_1)^2}{2\theta_2^2}} \quad H_1: \bar{H}_0$$

$$(-\infty; 1); [1; 2); \dots; [8; 9); [9; +\infty)$$

$$p_1 = \int_{-\infty}^1 p(x) dx \quad \dots \quad p_{10} = \int_9^{+\infty} p(x) dx$$

$$\text{ОМПГ: } L(\theta) = p_1^5 \cdot p_2^8 \cdot p_3^6 \cdot p_4^{12} \cdot p_5^{14} \cdot p_6^{18} \cdot p_7^{11} \cdot p_8^6 \cdot p_9^{13} \cdot p_{10}^7 = \max$$

$$\tilde{Q}_1 = 5,0291 \quad \tilde{Q}_2 = 3,5016$$

$$\tilde{D} = \sum_{i=1}^{10} \frac{(m_i - h \tilde{p}_i(\tilde{Q}))^2}{h \tilde{p}_i(\tilde{Q})} \approx 20,66$$

$$p\text{-value} = \int_{20,66}^{\infty} q(t) dt \approx 0,059 > \alpha \rightarrow \text{nen ver. omverwijzing } H_0.$$